

Ze Theory as an Interpretive Framework for Quantum Mechanics

Postulates, Formalism, and Open Problems

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Abstract

To present Ze Theory as an interpretive framework for quantum mechanics, reformulating known quantum phenomena in the language of Ze-observers—quantum systems by construction that possess proper time and act under conditions of struggle for existence. A Ze-observer is defined as a tuple $(H, \rho_Z, \{M_i\}, \tau_Z, \theta_Z)$, where H is a Hilbert space, ρ_Z is a belief state, $\{M_i\}$ is a POVM, τ_Z is a proper time counter, and θ_Z is a prediction threshold. The dynamics of τ_Z follow the rule: for a correct prediction (S-event), τ_Z is preserved; for an incorrect prediction (T-event), τ_Z decreases by 1. Two fundamental survival strategies of Ze are derived: internal adaptation (filtering of rare events) and external adaptation (niche construction). Within the proposed framework: (1) the arrow of time emerges from the decrease of τ_Z during T-events; (2) aging is interpreted as the dominance of the filtering strategy; (3) quantum entanglement corresponds to Ze generated by a common parent; (4) the double-slit experiment is reinterpreted through strategy choice; (5) spacetime is constructed from partial order and the commutation matrix of POVMs. Ze Theory offers a coherent interpretive framework for quantum mechanics, consistent with evolutionary epistemology. Open problems include deriving the Born rule from selection pressure, Lorentz invariance from Ze network dynamics, and the emergence of gravity.

Keywords: Ze Theory, interpretation of quantum mechanics, POVM, proper time, struggle for existence, collapse of the wave function, emergent spacetime.

1. Introduction

1.1. Motivation

Since the creation of quantum mechanics, the problem of interpreting its formalism has remained open. The Copenhagen interpretation, the Many-Worlds interpretation, de Broglie–Bohm theory, objective collapse models (GRW), the quantum Bayesian interpretation (QBism), and other approaches offer different ontologies, but none is universally accepted.

Ze Theory, developed by the author since 2024, offers a new perspective: quantum phenomena are reinterpreted in terms of Ze-observers—quantum systems by construction that possess proper time and act under conditions of struggle for existence.

1.2. Status of the Theory

Ze Theory is positioned as an interpretive framework, not as a derivation of quantum mechanics from more primitive primitives. It accepts the standard formalism of QM (Hilbert spaces, POVMs, Born rule) as axioms and proposes operational rules to which Ze-observers obey.

1.3. Structure of the Paper

Section 2 presents the five postulates of Ze Theory. Section 3 provides the mathematical formalism (POVM approach, Option D). Section 4 derives consequences from the postulates. Section 5 describes open problems. Section 6 contains discussion and conclusion.

2. Five Postulates of Ze Theory

Postulate 1 — Formal Primacy of Ze

In the axiomatics of Ze Theory, the concepts of time and space are not presupposed. They are derived:

- The time axis emerges as the average rate of T-events across a network of Ze-observers (a sequential, order-dependent channel)
- Space emerges as the set of pairs of Ze with commuting POVMs (a parallel, order-invariant channel)

Thus, Ze is logically (formally) primary with respect to spacetime. The statement that Ze exists "before time and space" (in a metaphysical sense) means this formal primacy, not chronological precedence before the Big Bang.

Falsifier: If it is shown that deriving the Minkowski metric or Lorentzian topology requires an a priori assumption of the existence of time or space, then the postulate of formal primacy of Ze will be refuted.

Mathematical justification: The derivation of the Minkowski metric from Ze dynamics is presented in the author's works (Tkemaladze, 2026).

Postulate 2 — Proper Time

Ze possesses proper time (τ_Z) when generated from another Ze. $\tau_Z \in \mathbb{N}$ is a discrete counter, inherited when a correlation is formed between Ze-observers.

Postulate 3 — Cost of Prediction

Ze predicts events in time:

- If the prediction is correct $\rightarrow$ proper time τ_Z is not spent (preserved)
- If the prediction is incorrect $\rightarrow$ proper time τ_Z decreases

Postulate 4 — Struggle for Existence (Two Survival Strategies)

In its struggle for existence, Ze develops two fundamental strategies:

- (a) Internal adaptation (filtering): Ze creates internal filters to ignore rare (hard-to-predict) events. This reduces the frequency of T-events and saves τ_Z . Cost: loss of information about the environment, reduced complexity.
- (b) External adaptation (niche construction): Ze actively modifies the environment it predicts, making its predictions self-fulfilling. This also reduces T-events but requires expenditure of τ_Z on action.

These strategies are analogous to the exploration/exploitation dilemma in adaptive systems theory (Mehlhorn et al., 2015).

Postulate 5 — Observation = Collapse

Ze changes the external environment by observation: Ze transitions matter from the wave phase to the material (classical) phase—exactly as in the double-slit experiment. The observation by Ze is the collapse of the wave function.

Distinction from the Copenhagen interpretation: In the standard Copenhagen interpretation, collapse is an epistemic update of the observer's information about the system. In Ze Theory, collapse is an ontological (real) physical event that occurs when Ze interacts with the environment. Ze does not merely learn the measurement result—it produces it.

3. Mathematical Formalism (POVM, Option D)

3.1. Definition of a Ze-Observer

A Ze-observer is defined as a tuple:

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Ze = (H,  $\rho_Z$ , {Mi},  $\tau_Z$ ,  $\theta_Z$ )  
where:  
H -- finite-dimensional Hilbert space ( $\dim H < \infty$ )  
 $\rho_Z \in \mathcal{D}(H)$  -- belief state of Ze (density matrix)
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$\{M_i\}_{i=1}^k$ -- POVM (positive operator-valued measure), $\sum M_i^\dagger M_i = I_H$
 $\tau_Z \in \mathbb{N}$ -- proper time counter (discrete, $\tau_Z(0) \geq 1$)
 $\theta_Z \geq 0$ -- prediction threshold

3.2. Dynamics of τ_Z

At each discrete time step t , the following process occurs:

1. Prediction: Ze computes the most likely outcome:

$$j^* = \operatorname{argmax}_j \operatorname{Tr}(M_j^\dagger M_j \rho_Z(t))$$

2. Realization: Outcome j occurs with probability:

$$p_j = \operatorname{Tr}(M_j^\dagger M_j \rho_Z(t))$$

3. Belief update (Lüders rule):

$$\rho_Z(t+1) = M_j \rho_Z(t) M_j^\dagger / \operatorname{Tr}(M_j \rho_Z(t) M_j^\dagger)$$

4. Proper time update:

- S-event ($j = j^*$): $\tau_Z(t+1) = \tau_Z(t)$ — time is not spent
- T-event ($j \neq j^*$): $\tau_Z(t+1) = \tau_Z(t) - 1$ — time decreases

5. Termination: If $\tau_Z = 0$, Ze ceases to exist.

3.3. Survival Strategies

Strategy A (internal adaptation / filtering):

After a T-event, Ze increases the prediction threshold:

$$\theta_Z(t+1) = \theta_Z(t) + \delta$$

A T-event is considered only for outcomes with $|p_j - p_{j^*}| > \theta_Z(t)$; otherwise, the event is classified as an S-event (even with formal mismatch).

Strategy B (external adaptation / niche construction):

Ze applies a unitary operator $U_Z(t)$ to the environment, maximizing the probability of its prediction:

$$U_Z(t) = \operatorname{argmax}_U \operatorname{Tr}(M_{\{j^*\}}^\dagger M_{\{j^*\}} \cdot U \rho_Z(t) U^\dagger)$$

3.4. Spacetime from a Ze Network

For a network of Ze-observers $\{Z_a\}$:

- Temporal structure: A partial order $<$ on the Ze network, defined as the transitive closure of causal connections (generation of one Ze by another).

- Spatial coordinates: Constructed from the commutation matrix:

$C(Z_a, Z_b) = ||[M^{\{a\}}, M^{\{b\}}]||$
 where $M^{\{a\}}$ and $M^{\{b\}}$ are the POVMs of the respective Ze. Spatial coordinates are obtained via multidimensional scaling (MDS). The connection to Minkowski geometry remains an open problem.

4. Consequences

From the formulated postulates and formalism, the following consequences are derived:

4.1. Arrow of Time

Ze generates the arrow of time: τ_Z decreases during T-events (incorrect predictions). The direction of time corresponds to the average decrease of τ_Z across the network.

4.2. Aging

Aging is interpreted as the dominance of Strategy A (filtering): $\theta_Z \rightarrow \infty$, Ze ceases to notice new (rare) phenomena, loses adaptability and informational complexity.

4.3. Death

Death of Ze occurs when $\tau_Z = 0$ —exhaustion of the proper time reserve.

4.4. Optimal State v^*

The optimal balance between T- and S-events is achieved at $v^* \approx 0.456$, corresponding to the maximum of informational complexity τ^{info} .

Note on two v values:* In passive mode (pure informational counter), the optimum is $1 - \ln 2 \approx 0.3069$. In active mode (Ze as an agent struggling for existence), the optimum shifts to ≈ 0.456 . The difference $\Delta v \approx 0.1491$ is interpreted as the "cost of agency."

4.5. Quantum Entanglement

Entangled Ze are those generated by a common parent, sharing the post-measurement state.

4.6. Double-Slit Experiment

Within Ze Theory, the double-slit experiment is reinterpreted:

- Strategy A (filtering): Ze does not distinguish which slit the particle passed through $\rightarrow$ interference pattern (wave phase)
- Strategy B (niche construction): Ze actively measures (interacts with the environment), recording the particle's path $\rightarrow$ collapse, disappearance of interference (material phase)

This corresponds to Postulate 5: observation by Ze is collapse.

5. Open Problems (Conjectures)

The following problems require further investigation:

#	Problem	Status
1	Born rule from selection pressure: Show that a population of Ze under selection for τ_Z converges to Born rule statistics	Conjecture
2	Lorentz invariance: Derive from Ze network dynamics (not from metric ansatz)	Conjecture
3	Gravity: Einstein equations as consistency condition of Ze network	Conjecture
4	Closed form of v^* : Find analytical expression for $v^* \approx 0.456$	Open question
5	Martingale hypothesis: Prove that $\tau_{\{Z_1\}} + \tau_{\{Z_2\}}$ is a martingale under mutual observation	Hypothesis
6	Collatz via Ze: Apply Ze potential $\Phi(n)$ and monotonicity of $\tau(n)$ to the Collatz conjecture	Application
7	Riemann zeros: Investigate $\zeta(s)$ zeros as fixed points of Ze flow on $\text{Re}(s)=1/2$	Application

6. Discussion and Conclusion

6.1. Comparison with Other Interpretations

Interpretation	Status of Collapse	Role of Observer	Proper Time
Copenhagen	Epistemic	Classical apparatus	No
Many-Worlds	None (branching)	Not needed	No
de Broglie-Bohm	None (pilot wave)	Not needed	No
GRW (objective collapse)	Ontological (spontaneous)	None	No
QBism	Epistemic (personalist)	Bayesian agent	No
Ze Theory	Ontological (observation = collapse)	Ze-observer	Yes (τ_Z)

6.2. Advantages of the Ze Approach

1. Unified language for describing quantum phenomena, aging, adaptation, and spacetime
2. Operationality of all key concepts (except Postulate 1, which is formal primacy)

3. Connection to evolutionary epistemology through survival strategies
4. Empirical consequences (via v^* , τ_Z dynamics, predictions for aging)

6.3. Limitations

1. Postulate 1 ("formal primacy") remains metaphysical in its phrasing, though operationalized
2. The Born rule and Lorentz invariance have not yet been derived from first principles
3. Experimental verification requires development of measurement protocols for τ_Z

6.4. Conclusion

Ze Theory offers a coherent interpretive framework for quantum mechanics, consistent with evolutionary epistemology and adaptive systems theory. It accepts the standard QM formalism as axiomatic and adds operational rules related to the proper time of observers and their struggle for existence. The main open problems—deriving the Born rule and Lorentz invariance—define the direction for future research.

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Conflict of Interest

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